ADEMP-Pre-Registration for Simulation Studies - Analysts level

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Template based on the paper:
Siepe, B. S., Bartoš, F., Morris, T. P., Boulesteix, A.-L., Heck, D. W., & Pawel, S. (2024). Simulation Studies for Methodological Research in Psychology: A Standardized Template for Planning, Preregistration, and Reporting. *Psychological Methods*. https://doi.org/10.1037/met0000695, https://doi.org/10.31234/osf.io/ufgy6

1 General Information

1.1 What is the title of the project?

Answer: Crowdsourcing methodological comparisons: A many-analysts simulation study to identify conditions under which results in convenience samples generalize to the target population.

1.2 Provide a description of the project.

Answer: We evaluate the performance of multiple effect size measures under non-representative sampling mechanisms. Our simulation study investigates how five commonly used estimators obtained from convenience samples generalize to the full population. To model non-representative sampling, we impose range restrictions on `X` and `Y` and compute each estimator from the resulting restricted samples. We then quantify relative bias and statistical power.

1.3 Did any of the contributors already conduct related simulation studies on this specific question?

Answer: No.

2 Aims

2.1 What is the aim of the simulation study?

Answer: Our goal is to evaluate the extent to which common effect size measures generalize from convenience samples to the full population when sampling is distorted by range restriction in either `X` or `Y`. We assess this by quantifying the relative bias and power of five estimators under range restriction.

3 Data-Generating Process

3.1 How will the parameters for the data-generating process (DGP) be specified?

Answer: The DGP includes a numeric outcome `Y`, and a continuous or a binary predictor of interest `X`. The relationship between `X` and `Y` is linear and additive, without interactions or nonlinear terms. Errors will be independent and homoscedastic:

$$\varepsilon \sim \mathcal{N}(0, \sigma^2)$$

A minimal example is:

$$y = \beta_0 + \beta_1 x + \varepsilon$$

Additional covariates (confounders correlated with `X`) can be included and `X` can be sampled from a normal/uniform/empirical/... distribution. Furthermore, sampling mechanisms that systematically alter the range or distribution of either `X` or `Y` will be implemented. We may specify two distinct DGPs: one for a numeric predictor `X` and another for a binary predictor `Y` in separate files.

3.2 What will be the different factors of the data-generating mechanism?

Answer: The following factors will be varied:

- the true effect of `X` on `Y` (β_1) (with three levels: weak, moderate and strong)
- the strength of the range restriction applied to `X` and/or `Y` (with four levels: none, weak, moderate and strong)

3.3 If possible, provide specific factor values for the DGP as well as additional simulation settings.

Answer: We will use the following values for our DGP:

- β_1 = (NA, NA, NA); replace NAs!
- restrict = ("none", "weak", "moderate", "strong")

- n: Number of observations in each sample

The sampling mechanisms which lead to the restrictions in `X` and `Y` are [probabilist/deterministic]. For the other DGP components (e.g., sample size, distributional form of `X`, error variance) we will use the following values/definition: ... We define a `dgp()` function, which returns a list with two data frames: `convenience_x` and `convenience_y`, respectively. If `restrict = "none"`, i.e., when the argument `"none"` is passed to `restrict`, no restriction is applied. In this case, `convenience_x` and `convenience_y` both contain the same unrestricted dataset. The model formula is `Y` $\sim$ `X` and we will run a validation function to verify that the constraints for the confirmatory part are satisfied and the estimator for β_1 is unbiased in the unrestricted sample.

3.4 If there is more than one factor: How will the factor levels be combined and how many simulation conditions will this create?

Answer: We will vary the conditions in a fully factorial manner. This will result in 3 (effect sizes (β_1) $\times$ 4 (restriction strength) $\times$ 2 (restriction on X or Y) = 18 simulation conditions. For each of the conditions, five effect sizes measures will be derived. For more details see next section.

4 Estimands and Targets

4.1 What will be the estimands and/or targets of the simulation study?

Answer: In this simulation study, we consider five effect size measures of interest:

- the unstandardized regression coefficient (β_1),
- the standardized regression coefficient (β_1^{std}),
- the coefficient of determination (R^2),
- the Pearson correlation between `X` and `Y` (r),
- Cohen's d.

5 Methods

5.1 How many and which methods will be included and which quantities will be extracted?

Answer:

Effect sizes. The core estimation procedure is an ordinary least squares (OLS) regression of `Y` on `X` (and any additional covariates specified in the DGP), implemented via `stats::lm()` in R with default settings. Depending on the type of predictor `X`, the script will proceed as follows:

- For **numeric** `X`, a linear regression model

  $$Y \sim X \text{ (+ additional covariates)}$$

  will be fitted using `lm()` to derive the estimated unstandardized regression coefficient for β_1. In addition, a standardized version of this model will be fitted after `Y` and the predictors are z-standardized within the script to derive an estimate for the standardized regression coefficient β_1^{std}. Furthermore, an estimate for the partial coefficient of determination R^2 is computed as:

  $$\hat{R^2} = \frac{\hat{\beta_1}^2 \widehat{Var(X)}}{\widehat{Var(Y)}}$$

  with $\hat{\beta_1}$ denotes the slope estimate from the fitted linear model and $\widehat{Var(X)}$ and $\widehat{Var(Y)}$ denote the empirical variances computed from the simulated dataset. In addition, the Pearson correlation coefficient r is estimated as:

  $$\hat{r} = \frac{\widehat{Cov(X,Y)}}{\sqrt{\widehat{Var(X)}}\sqrt{\widehat{Var(Y)}}} = \frac{\hat{\beta_1}\sqrt{\widehat{Var(X)}}}{\sqrt{\widehat{Var(Y)}}}$$

  implying that $r = sign(\hat{\beta_1})\sqrt{\hat{R^2}}$.
  Note: In the univariate case, r and R^2 coincide with the standardized regression coefficient. In multivariate models, however, the r does not represent a partial effect, as it does not adjust for the presence of additional predictors.

- For **binary** `X` (two groups), the same linear model and formulas will be used to derive β_1, β_1^{std} and R^2. Instead of r the Cohen's d will be computed as:
$$\hat{d} = \frac{\hat{\bar{Y}}_1 - \hat{\bar{Y}}_0}{\hat{s}}$$
with
$$\hat{s} = \sqrt{\frac{(n_0 - 1)\hat{s_0}^2 + (n_1 - 1)\hat{s_1}^2}{n_0 + n_1 - 2}}),$$
with $\hat{\bar{Y}}_0$ and $\hat{\bar{Y}}_1$ denoting the group means, n_0 and n_1 group sizes, and $\hat{s_0}^2$ and $\hat{s_1}^2$ group variances in the simulated datasets.

From each fitted model and for each generated sample (convenience samples restricted on `X` or `Y`, and matched-size random samples), the script will automatically extract the following quantities, depending on the type of `X`:

- the unstandardized regression coefficient for `X` ($\hat{\beta}_1$), its standard error, and a two-sided 95% confidence interval;
- the standardized regression coefficient ($\hat{\beta}_1^{std}$), its standard error, and a 95% confidence interval;
- the coefficient of determination $\hat{R}^2$;
- for numeric `X`, the Pearson correlation $\hat{r}$ between `X` and `Y`, including its 95% confidence interval;
- for binary `X`, Cohen's $\hat{d}$ and its 95% confidence interval.

Computation of true values. For β_1 the true value is known from the DGP and is therefore set directly to this value.
β_1^{std} is computed as
$$\beta_1^{std} = (R_{XX}^{-1} r_{XY})_1,$$
where R_{XX} denotes the population correlation matrix of the predictors and r_{XY} the vector of correlations between each predictor and the outcome. The subscript $(\cdot)_1$ refers to the element corresponding to the focal predictor.
The Pearson correlation coefficient r is computed as
$$r = \beta_1 \frac{\sqrt{Var(X)}}{\sqrt{Var(Y)}},$$

and the coefficient of determination R^2 is computed as

$$R^2 = \beta_1^2 \frac{Var(X)}{Var(Y)}$$

$Var(X)$, $Var(Y)$, R_{XX} and r_{XY} are obtained as Monte-Carlo averages of the empirical variances computed from 1,000 independent simulated datasets generated under the unrestricted data-generating process.
If `X` is binary, Cohen's d is computed as

$$d = \frac{\mu_1 - \mu_0}{s_p},$$

where μ_1 and μ_0 denote the population means of `Y` in the two groups and s_p is the pooled standard deviation of `Y` across groups. These quantities are approximated by Monte Carlo averages across 1,000 independent simulated datasets.

Coverage, power, and power loss. For each effect size measure θ, except R^2, and each simulation i, the script will compute a two-sided 95% confidence interval $[L_i, U_i]$ for the parameter of interest (for regression coefficients, Pearson r, and Cohen's d). Coverage will be assessed by the indicator

$$C_i = \mathbb{I}\{\theta \in [L_i, U_i]\},$$

and the empirical coverage probability will be estimated by

$$\widehat{\text{Cov}} = \frac{1}{n_{\text{sim}}} \sum_{i=1}^{n_{\text{sim}}} C_i.$$

Using the same confidence intervals, we define rejection of the null hypothesis of no effect by

$$R_i = \mathbb{I}\{0 \notin [L_i, U_i]\}.$$

In conditions where the true effect is zero, the mean of R_i estimates the type I error rate; in conditions with a non-zero true effect, it estimates power. The script will compute these rejection rates both for the convenience samples and for matched-size random samples.

6 Performance Measures

6.1 Which performance measures will be used?

Answer: Performance will be evaluated using the *relative bias* and the statistical power. For each effect size measure, relative bias will be summarized across simulation conditions using the mean and Monte Carlo standard errors.
Let θ denote the true value of a given effect size and $\hat{\theta}_i$ the corresponding estimate in simulation i, for $i = 1, \ldots, n_{\text{sim}}$.

Relative bias. For $\theta \neq 0$, we define the relative bias in simulation i as

$$b_i = \frac{\hat{\theta}_i - \theta}{\theta},$$

and summarize it across replications by the mean relative bias

$$\widehat{\text{RelBias}} = \frac{1}{n_{\text{sim}}} \sum_{i=1}^{n_{\text{sim}}} b_i.$$

These quantities will be reported as percentages ($\times 100$) for ease of interpretation.
The Monte Carlo standard error (MCSE) of the mean relative bias will be computed as

$$\text{MCSE}_{\widehat{\text{RelBias}}} = \frac{S_b}{\sqrt{n_{\text{sim}}}},$$

where S_b is the sample standard deviation of $(b_1, \ldots, b_{n_{\text{sim}}})$. We will also report empirical 2.5% and 97.5% quantiles of the b_i as a Monte Carlo interval for the distribution of relative bias.

6.2 How will Monte Carlo uncertainty of the estimated performance measures be calculated and reported?

Answer: For each simulation condition, we will compute and plot MCSEs for all performance measures.

6.3 How many simulation repetitions will be used for each condition?

Answer: We will perform $n_{\text{sim}} = 10{,}000$ repetitions per condition (for each combination of DGP, effect size measure, restriction level, true effect size level, and sample type).
This choice is guided by standard recommendations for simulation studies and by the goal of keeping Monte Carlo uncertainty negligibly small for our primary performance measures.

6.4 How will missing values due to non-convergence or other reasons be handled?

Answer: Missing values and non-convergence is not expected. Missing values will be coded as NA and will not be imputed.

6.5 How do you plan on interpreting the performance measures? (optional)

Answer: We will interpret performance primarily in terms of the magnitude and direction of the relative bias for each effect size measure across simulation conditions. Because our goals are descriptive rather than inferential, we do not define strict thresholds for "acceptable" or "unacceptable" levels of bias. Power will be visualized in a plot for each effect size β_1 and restriction strength. The absolute difference between the power estimates represents the loss in power and can be interpreted directly from the graph.

Interpretation across restriction levels and effect sizes We will compare the mean relative bias and power across the three predefined levels of range restriction (weak, moderate, strong) and across the three true effect-size levels (β_1: weak, moderate, strong). A monotonic increase in bias with stronger restriction will be interpreted as evidence that increasing restriction systematically degrades estimator performance. Likewise, systematic decreases in power across restriction levels will be interpreted as evidence of power loss due to stronger restriction. The same comparisons will be made across effect-size levels to assess whether estimator performance varies with the underlying true effect strength.

7 Other

7.1 Which statistical software/packages do you plan to use?

Answer: All simulations and analyses will be conducted in `R` (R Core Team, 2025).
The core simulation wrapper and analysis scripts rely on base `R` and the following packages in their CRAN versions current at the time of running the study:

- `stats` (R Core Team, 2025) for model fitting and inferential procedures (e.g., `lm()` for linear regression);
- `ggplot2` (Wickham, 2016) for data visualization;
- `dplyr` (Wickham et al., 2023) for data manipulation.

If additional packages are required for the DGP (e.g., for multivariate data generation), these will be loaded within the corresponding DGP script and documented.

7.2 Which other steps will you undertake to make simulation results reproducible? (optional)

Answer: We will produce a text file including the `sessionInfo()` output.

7.3 Is there anything else you want to preregister? (optional)

Answer: No.

References

R Core Team. (2025). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. Vienna, Austria. https://www.R-project.org/

Wickham, H. (2016). *Ggplot2: Elegant graphics for data analysis*. Springer-Verlag New York. https://ggplot2.tidyverse.org

Wickham, H., François, R., Henry, L., Müller, K., & Vaughan, D. (2023). *Dplyr: A grammar of data manipulation* [R package version 1.1.4]. https://doi.org/https://doi.org/10.32614/CRAN.package.dplyr